AI-Powered Insurance Claims Intelligence Assistant

Overview

Insurance companies process thousands of claims daily involving vehicle accidents, health reimbursements, property damages, and policy settlements. Claims analysts spend significant time reviewing claim documents, customer records, policy conditions, historical claims, and fraud indicators to validate claims and detect suspicious activities.

Claims teams frequently investigate situations such as:

- "Multiple claims submitted for similar vehicle damage patterns within a short duration."
- "Customer claim history shows unusual frequency increases."
- "Repair cost estimates significantly exceed historical averages."
- "Policy mismatch causing repeated claim rejections."
- "Potential fraud indicators detected across geographically distributed claims."

Traditional claims processing systems rely heavily on manual verification and rule-based fraud detection, increasing processing time and operational costs.

The system should help insurance operations teams retrieve historical claims, identify fraud risks, recommend investigation actions, and generate explainable insurance intelligence.

Objective

The objective of this project is to build an AI-powered insurance claims intelligence assistant that enables insurance teams to investigate claims using natural-language queries and intelligent fraud analysis.

The system should retrieve historical claim incidents, identify suspicious claim behavior, correlate fraud signals, and generate explainable risk insights to improve operational efficiency and fraud prevention.

Key Capabilities

Natural Language Claims Queries

Claims analysts can describe insurance issues conversationally.

Semantic Claim Retrieval

Retrieve historical insurance claims using conceptual similarity rather than exact keywords.

Fraud Pattern Analysis

Identify suspicious claim behaviors using historical fraud patterns.

Policy Validation Intelligence

Validate policy compliance and eligibility conditions.

Risk Scoring

Estimate fraud probability and investigation priority.

Insurance Knowledge Summarization

Generate summaries of operational claim trends and fraud patterns.

Explainable Recommendations

Provide grounded investigation recommendations.

Multi-System Understanding

Support claims systems, policy systems, fraud monitoring systems, and customer information platforms.

Requirement 1 (Basic)

- Basic RAG for claim retrieval
- Hybrid search (keyword + semantic)
- Claims semantic search
- Metadata filtering (claim type, policy category, region)
- Basic fraud recommendation engine
- Input validation guardrails
- Incident similarity ranking
- Investigation recommendation generation

Expose the core functionality through API endpoints so users or other applications can interact with the system.

Requirement 2 (Advanced)

- DeepEval for maintenance recommendation quality evaluation
- Equipment anomaly correlation analysis
- Reranking using maintenance embeddings
- LLM-as-judge for maintenance validation
- Token optimization for large-scale telemetry analysis

Multi-Agent Equipment Intelligence System

- Equipment Retrieval Agent
- Reliability Analysis Agent
- Maintenance Agent

    - Recommendation Agent

- Fraud prediction intelligence
- Claims dashboard
- Agent-to-Agent (A2A) communication for escalation workflows

Build a simple front-end interface to demonstrate how biomedical engineers can interact with the system.

Additional Capabilities (Optional)

- Fraud trend monitoring
- Cross-region fraud pattern analysis
- Claim investigation prioritization intelligence
- Customer behavioral risk profiling
- Feedback loop for fraud model improvement
- Real-time insurance analytics dashboard
- Integration with claims management systems
- Automated claim anomaly alerts

Deliverables

1. Architecture Diagram

Include a high-level view showing:

- claims ingestion pipeline
- fraud event processing
- document chunking and embedding generation
- hybrid retrieval layer
- multi-agent fraud intelligence pipeline
- recommendation generation

2. Design

Explain system design decisions and trade-offs such as:

- vector database selection
- chunking strategy for claim documents
- hybrid search vs semantic-only retrieval
- agent orchestration architecture

- fraud correlation strategy
- operational reliability guardrails

3. Full Executable Code (Microservice)

Focus on modularity and clarity.

Provide a README explaining:

- Project setup
- Claims data ingestion process
- Sample fraud investigation query
- Example recommendation generation
- API usage examples

4. Panel Presentation (10 Minutes)

Demo Flow

- Claims analyst enters insurance investigation query
- System retrieves related historical claims
- Multi-agent analysis identifies probable fraud risks
- System generates explainable investigation guidance

Q&A with panel (2 minutes)

Dataset Name

Vehicle Insurance Claim Fraud Detection Dataset : **fraud_oracle.csv**

Primary Link

Vehicle Claim Fraud Dataset

Alternative Links

Insurance Claim Prediction Dataset

Format

CSV

Key Fields

claim_id, policy_type, accident_type, claim_amount, customer_region, fraud_label, incident_date, customer_history, claim_status

Reason

Insurance operational datasets containing claim histories, fraud indicators, policy information, customer behavioral patterns, and operational risk signals suitable for semantic retrieval, anomaly detection, fraud intelligence, predictive risk analysis, and AI-powered insurance claims monitoring systems.